

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2022/2023

MAY/JUNE EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

TUESDAY, 30 MAY 2023

TIME: 6.30 PM – 8.30 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR OF SCIENCE (HONOURS) IN MANAGEMENT MATHEMATICS WITH
COMPUTING

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

- a) You, as an appointed software engineer are in-charge for designing a web-based system for Zara hairstyling Academy. Currently the management has been using manual way to manage their academy and now has decided to automate all their registration, classes scheduling, attendance, view result, and payment processes. Design **FOUR (4)** sets of aperiodic stimuli-response for Zara hairstyling academy web-based system as mentioned above. (8 marks)
- b) VAC.com is an online website which allows the users to book and rent cars for vacation. The company's service areas include the whole country of Malaysia. The website allows users to view available car models, rental fee, make booking, and make payment. A booking number will be generated once the payment is confirmed. Recently VAC.com is planning to extend their website on mobile platform as mobile users are increasing dramatically.

Propose **FIVE (5)** user interface design principles for VAC.com mobile application. (5 marks)

- c) Identify **TWO (2)** functional and **TWO (2)** non-functional requirements as shown in Figure 1. (4 marks)

This e-shopping system shall allow online customers to search, select, order and make payment for their desired product.
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Figure 1: Statement

- d) Compare and contrast between Perfective Maintenance and Adaptive Maintenance. Which maintenance is invoked in improving a mobile application with a new login feature? Justify your answer. (8 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 2**

- a) You, as a project manager, have been assigned to lead a project team to develop an online inventory system for Sri Ternak wholesale department. Currently, they have 3 branches located in Perak, Penang and Selangor. The system will allow the staff to check on the current stock, send a notification if an item is out of stock, print the monthly stock details, manage the orders and delivery. The management ensure that all the 3 branches should have a proper data sharing to avoid any issues in future.

Recommend and illustrate an appropriate system organisation model for the above project with justification. (10 marks)

- b) Managing a project group is one of the challenging and critical management tasks in software development projects. Analyse **FOUR (4)** factors which influence the project group. (8 marks)
- c) System requirements specification (SRS) is the main output document of *Requirements Specification* stage and it is usually presented in a structured form of natural language supplemented by system models and tables.

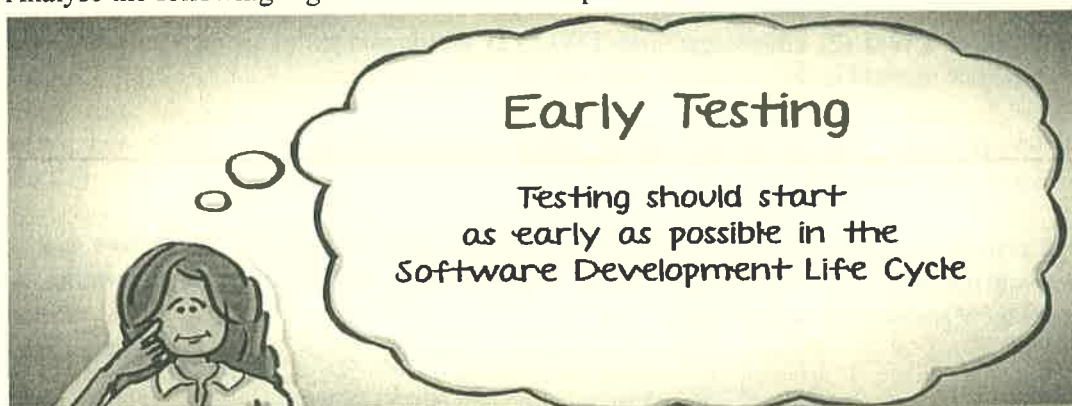
(i) Analyse **THREE (3)** possible problems of writing this document in natural language. (3 marks)

(ii) Explain **FOUR (4)** alternatives to using natural languages. (4 marks)

[Total: 25 marks]

Question 3

- a) Analyse the following Figure 2 and answer the questions below.



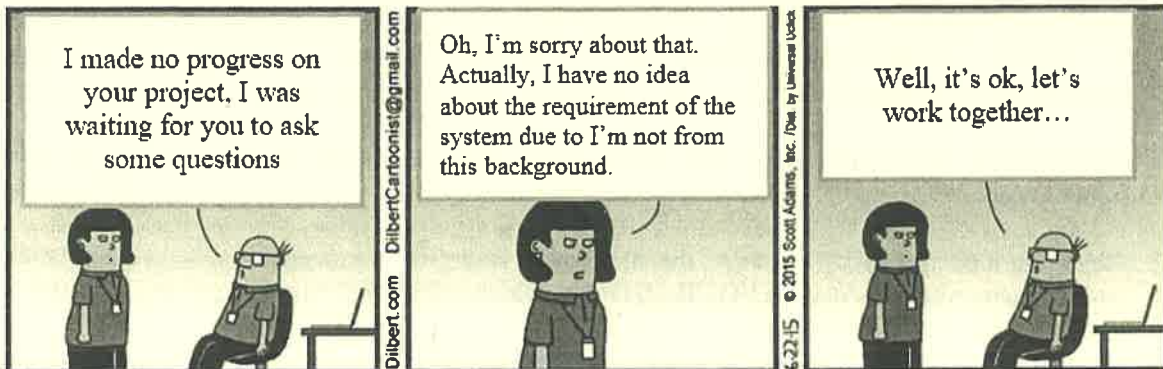
Source: <http://www.guru99.com/software-testing>

Figure 2: Statement

- (i) Do you agree with the above statement? Provide **THREE (3)** justification about the importance of software testing in your answer. (6 marks)
- (ii) Identify any **FIVE (5)** stages of software testing process. (10 marks)

BACS2163 SOFTWARE ENGINEERING**Question 3 (Continued)**

- b) Analyse the following scenario in Figure 3 and recommend with justification an appropriate software process model to complete the project. (9 marks)



Source: <https://www.liamwho.com/software-process-models-w02/>

Figure 3: Scenario

[Total: 25 marks]

Question 4

- a) Divo Company is developing a software for online food ordering system for a restaurant. Among the requirements of the system are view the menu and ingredients, select the menu, make payment, cancel order, able to read and post new reviews, contact the restaurant via WhatsApp or call, track their orders. Divo has decided to apply viewpoint-oriented analysis in the requirement analysis stage.
- Discuss **TWO (2)** advantages and **TWO (2)** disadvantages of using viewpoint-oriented analysis method in the requirement analysis. (4 marks)
 - Identify any **SIX (6)** viewpoints of above online food ordering system and prepare a viewpoint hierarchy. (12 marks)
- b) In a project management, a project manager would need to consider the factors that could motivate employees throughout the project. Suggest and explain **THREE (3)** motivating factors based on Maslow motivation model. (9 marks)

[Total: 25 marks]